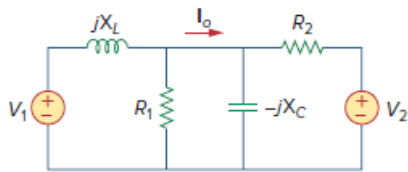


Problem

Using Fig. 10.87, design a problem to help other students better understand the superposition theorem.

Figure 10.87:



Step-by-step solution

Step 1 of 11

Consider the following circuit diagram:

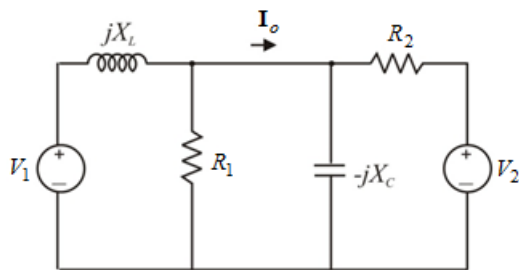


Figure 1

Step 2 of 11

Assume the network parameter values and consider the following circuit:

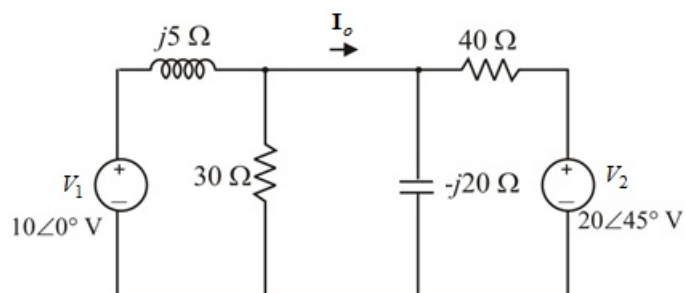


Figure 2

Step 3 of 11

Assume that, $\mathbf{I}_o = \mathbf{I}'_o + \mathbf{I}''_o$.

According to superposition theorem $\mathbf{I}'_o$ and $\mathbf{I}''_o$ are the currents due to the voltage sources $10\angle 0^\circ \text{ V}$ and $20\angle 45^\circ \text{ V}$ respectively.

Consider the following circuit in which $20\angle 45^\circ \text{ V}$ voltage source is kept short circuit to find $\mathbf{I}'_o$:

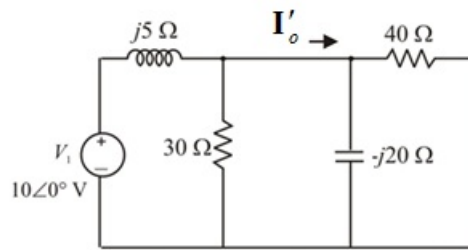


Figure 3

Step 4 of 11

Calculate the value of $\mathbf{Z}_1$.

$$\begin{aligned}\mathbf{Z}_1 &= (-j20 \, \Omega) \parallel (40 \, \Omega) \\ &= \frac{(-j20)(40)}{-j20 + 40} \\ &= 8 - j16 \, \Omega\end{aligned}$$

Transform the voltage source to current source:

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{V}_1}{j5 \, \Omega} \\ &= \frac{10\angle 0^\circ \text{ V}}{j5} \\ &= -j2 \text{ A}\end{aligned}$$

Step 5 of 11

Calculate the value of $\mathbf{Z}_2$:

$$\begin{aligned}\mathbf{Z}_2 &= (j5 \, \Omega) \parallel (30 \, \Omega) \\ &= \frac{(j5)(30)}{j5 + 30} \\ &= 0.81 + j4.86 \, \Omega\end{aligned}$$

Modify the circuit as follows:

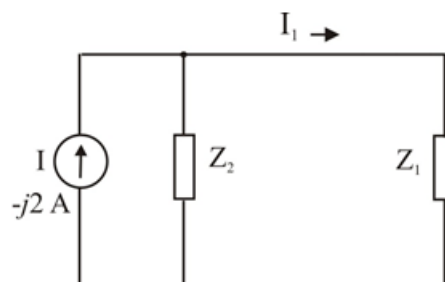


Figure 4

Step 6 of 11

Use current division method to calculate the value of $\mathbf{I}'_o$:

$$\begin{aligned}\mathbf{I}'_o &= \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} (\mathbf{I}) \\ &= \frac{0.81 + j4.86}{8 - j16 + 0.81 + j4.86} (-j2) \\ &= 0.5139 + j0.466 \text{ A}\end{aligned}$$

Therefore, the value of $\mathbf{I}'_o$ is $0.5139 + j0.466 \text{ A}$.

Step 7 of 11

Consider the following circuit in which $10\angle 0^\circ \text{ V}$ voltage source is kept short circuit to find $\mathbf{I}''_o$:

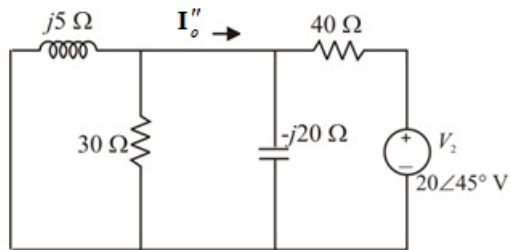


Figure 5

Step 8 of 11

Transform the voltage source to current source:

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{V}_2}{40 \Omega} \\ &= \frac{20\angle 45^\circ \text{ V}}{40} \\ &= 0.5\angle 45^\circ \text{ A}\end{aligned}$$

Step 9 of 11

Modify the circuit as follows:

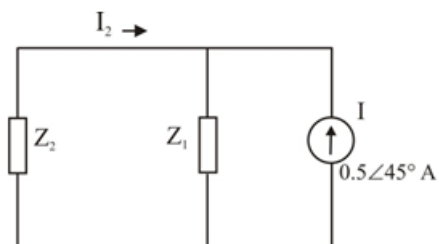


Figure 6

Step 10 of 11

Use current division method to calculate the value of $\mathbf{I''}$:

$$\begin{aligned}\mathbf{I''} &= \frac{-\mathbf{Z_1}}{\mathbf{Z_1} + \mathbf{Z_2}} (\mathbf{I}) \\ &= \frac{8 - j16}{8 - j16 + 0.81 + j4.86} (0.5 \angle 45^\circ) \\ &= -0.5268 - j0.345 \text{ A}\end{aligned}$$

Therefore, the value of $\mathbf{I''}$ is $-0.5268 - j0.345 \text{ A}$.

Step 11 of 11

Calculate the value of required current:

$$\begin{aligned}\mathbf{I_o} &= \mathbf{I'_o} + \mathbf{I''_o} \\ &= 0.5139 + j0.466 \text{ A} - 0.5268 - j0.345 \\ &= -0.0129 + j0.12 \\ &= 0.1216 \angle 96.08^\circ \text{ A}\end{aligned}$$

Hence, the current value is calculated by using the superposition theorem and the value is

$$\boxed{0.1216 \angle 96.08^\circ \text{ A}}.$$